Midterm Review Tutorial

STAT3612: Statistical Machine Learning

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Outline for Today

1. Exam Overview & Core Concepts
   - Fundamental Distinctions
   - Optimization & The Bias-Variance Tradeoff
2. Linear Models
   - Linear & Polynomial Regression
   - Regularization: Ridge, LASSO & Elastic Net
3. Classification Models
   - Evaluating & Building Classifiers
   - Non-Parametric Methods & Classifier Comparisons
4. Tree-Based Methods & SVM
   - Decision Trees & Ensembles
   - Support Vector Machines (SVM)
5. Final Advice & Q&A

What to Expect on the Midterm

The exam will test your understanding at three levels:

1. **Conceptual Understanding (True/False & Multiple Choice)**
   - Requires precise definitions. Pay close attention to words like "always," "never," or "only."
   - Watch out for negative marking on True/False (± 2 marks)—don't guess blindly!
2. **Explanatory Ability (Short Answer)**
   - Explain core principles (e.g., bias–variance tradeoff, learning rate, validation strategies).
   - Compare and contrast models (e.g., LDA vs. QDA, Ridge vs. LASSO).
3. **Application & Calculation (Applied Problems)**
   - Given a small dataset, fit a linear model, draw a decision boundary, or compute a split.
   - Match models to training errors or regularization paths with clear justification.

Our Goal Today

Connect the concepts (the "why") to the exam problem-solving (the "how").

Concept 1: Regression vs. Classification

Always start by checking the type of the **output (target) variable** Y:

Regression

- Output (Y) is quantitative (numerical).
- Examples: House price, temperature, visitor counts.

Classification

- Output (Y) is qualitative (categorical).
- Examples: Spam/Ham, Disease status, Digit 0–9.

Exam Questions (Practice Paper 1 Q1.1, Midterm 2022 Q1.1, Practice Paper 2 Q1.1)

1. *"A company wants to predict the **number of visitors** to a web page. Since the number of visitors is an integer, this is best viewed as a classification problem."*
2. *"For classification, the input variables X have to be categorical." / "Linear regression requires all input variables X to be numerical."*

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   FALSE: A count is a quantitative measurement → **Regression** (e.g., Poisson or linear regression).
2. *"For classification, the input variables X have to be categorical."* / *"Linear regression requires all input variables X to be numerical."*

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   FALSE: A count is a quantitative measurement → **Regression** (e.g., Poisson or linear regression).
2. *"For classification, the input variables X have to be categorical."* / *"Linear regression requires all input variables X to be numerical."*
   BOTH FALSE: The distinction depends **only on the output** Y; inputs X can be numerical or categorical (via one-hot encoding).

Concept 2: Parametric vs. Non-Parametric Models

How much structure do we assume about the unknown function $f(X)$ before seeing the data?

Parametric Models

- Assume a specific functional form for $f(X)$ (e.g., linear: $f(x) = \theta_0 + \theta_1 x_1 + \cdots + \theta_p x_p$).
- Learning reduces to estimating a fixed, finite set of parameters $\boldsymbol{\theta} = (\theta_0, \ldots, \theta_p)^T$.
- The number of parameters is fixed in advance and **does not grow** with sample size n.
- **Pros:** Easier to fit and interpret; needs less training data.
- **Cons:** Higher bias if the assumed form is far from the true $f(X)$.

Non-Parametric Models

- Make no explicit assumption about the global functional form of $f(X)$.
- The model structure adapts to the data; effective complexity can grow with sample size n.
- Predictions are driven directly by local training observations or data-driven partitions.
- **Pros:** Highly flexible; can capture complex non-linear patterns (low bias).
- **Cons:** Requires more data; prone to high variance (overfitting).

Parametric vs. Non-Parametric: Model Map & Exam Trap

Category	Models in STAT3612	What is learned?	Flexibility Control
Parametric	Linear / Polynomial Reg. Ridge / LASSO / Elastic Net Logistic Reg. / LDA / QDA Linear SVM	Fixed vector $\hat{\boldsymbol{\theta}}$ Shrunk vector $\hat{\boldsymbol{\theta}}$ $\hat{\boldsymbol{\theta}}$ or $(\hat{\pi}_k, \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k)$ Hyperplane $(\hat{\mathbf{w}}, \hat{b})$	Degree d Penalty λ (or α) Feature choice / $\boldsymbol{\Sigma}_k$ Cost C
Non-Parametric	k-Nearest Neighbors (kNN) Decision Trees / RF / Boosting Kernel SVM (e.g., RBF kernel)	Stores all n training points Data-driven split rules Support vectors & dual $\hat{\alpha}_i$	Neighborhood size K Tree depth / pruning Cost C, bandwidth γ

Exam Question (Practice Paper 1, Q1.4)

*"A non-parametric model for classification **always** achieves zero training error since the complexity grows with data."*

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FALSE. Non-parametric models still have hyperparameters that control smoothness! For example, k**NN with** $K = 10$ (or a shallow Decision Tree) averages over multiple points and will **not** achieve 0% training error when classes overlap.

Concept 3: Generative vs. Discriminative Models

Two fundamentally different approaches to building a classifier:

- **Generative Models (LDA, QDA, Naive Bayes)**
  - Model how each class generates features: estimate class-conditional density $P(X \mid Y = k)$ and class prior $P(Y = k)$, i.e., the joint distribution $P(X, Y)$.
  - Ask: *"Which class is most likely to have generated this data point?"* and use **Bayes' theorem** to get $P(Y = k \mid X)$.
- **Discriminative Models (Logistic Regression, SVM, Decision Trees, kNN)**
  - Directly model the posterior $P(Y = k \mid X)$ or the decision boundary.
  - Ask: *"Which side of the decision boundary does this point fall on?"* without modeling how X is distributed.

Exam Questions (Practice Paper 1 Q2.1, Midterm 2022 Q1.3)

1. **Q2.1:** *Explain the difference between a generative and a discriminative model in one sentence.*

2. **Midterm 2022 Q1.3:** *"k-nearest neighbors is a generative model."*

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1. **Q2.1:** *Explain the difference between a generative and a discriminative model in one sentence.*
   Answer: Generative models learn the joint distribution $P(X, Y)$ (via $P(X \mid Y)$ and $P(Y)$), whereas discriminative models directly learn $P(Y \mid X)$ or the decision boundary.
2. **Midterm 2022 Q1.3:** *"k-nearest neighbors is a generative model."*

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2. **Midterm 2022 Q1.3:** *"k-nearest neighbors is a generative model."* $\rightarrow$ **FALSE** (it directly estimates $P(Y \mid X)$ locally without modeling $P(X \mid Y)$).

Concept 4: Optimization with Gradient Descent & SGD

Many models (e.g., Linear/Logistic Regression) are trained by minimizing a cost function $J(\boldsymbol{\theta})$. Stochastic Gradient Descent (SGD) updates parameters iteratively using mini-batches:

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla J_i(\boldsymbol{\theta}^{(t)})$$

where the step size / learning rate η (Practice Paper 2, Q2.2) controls how big a step we take.

Step Size Too Large (Practice 1, Q2.2)

*What happens if η is set **too large**?*

Step Size Too Small (Practice 1, Q2.3)

*What happens if η is **too small** (or decays too fast)?*

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*What happens if η is set **too large**?*

- **Macro effect:** Updates overshoot the minimum.
- **Micro behavior:** Iterates oscillate wildly or diverge completely (loss explodes).

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Step Size Too Small (Practice 1, Q2.3)

*What happens if η is **too small** (or decays too fast)?*

- **Macro effect:** Convergence becomes extremely slow.
- **Micro behavior:** May never reach the optimum in practical time (or get trapped in a poor local minimum).

Concept 5: The Bias-Variance Tradeoff

The fundamental tension when choosing model complexity/flexibility:

Bias

- Error from overly restrictive model assumptions.
- High bias $\rightarrow$ Underfitting.
- Example: Fitting a straight line to a curved relationship.

Variance

- Error from excessive sensitivity to training set fluctuations.
- High variance $\rightarrow$ Overfitting.
- Example: A high-degree polynomial fitting random noise.

The Core Decomposition

As model flexibility increases: **Training Error** $\downarrow$, **Bias** $\downarrow$, but **Variance** $\uparrow$.

$$\text{Expected Test MSE} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

Our goal is to find the **sweet spot** in flexibility that minimizes total test error.

Bias-Variance in Action: Exam Questions

True/False Questions (Practice Paper 1 Q1.8, Midterm 2022 Q1.4 & Q1.5)

1. *"A model with lower bias always performs better than a model with higher bias in terms of test MSE."*

2. *"The training error usually decreases when we increase model flexibility."*

Short Answer (Midterm 2022, Q2.3)

*"Explain how you can tune each model in order to **decrease bias** (i.e., increase flexibility):"*
Target Models: **(a)** Polynomial Regression (degree d), **(b)** kNN (neighbors K), **(c)** LASSO (penalty λ).

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- **(a) Polynomial Regression:** Increase the polynomial degree d (captures higher-order curvature).
- **(b) k-Nearest Neighbors (kNN):** Decrease K (smaller neighborhood $\rightarrow$ lower bias, higher variance).
- **(c) LASSO Regression:** Decrease the penalty parameter λ (less shrinkage allows a more flexible fit).

Concept 6: Model Assessment (Validation Set vs. k-Fold CV)

How do we estimate test error to select hyperparameters (such as λ, K, or degree d)?

Validation Set Approach

- Randomly split data once into training and validation sets.
- **Pros:** Simple and computationally fast.
- **Two Drawbacks (Midterm 2022, Q2.2):**
  1. High variability: The validation error estimate depends heavily on which specific observations land in the split.
  2. Overestimates test error (upward bias): Fewer observations are available for training, so the fitted model performs worse than one trained on the full dataset.

k-Fold Cross-Validation

- Partition data into k equal folds; train on $k-1$ folds and validate on the held-out fold, repeating k times.
- **Pros:**
  1. Lower variability because we average validation error over all k folds.
  2. Less upward bias because each fit uses $\frac{k-1}{k}$ (nearly all) of the data for training.
- **Cons:** Requires fitting the model k times.

Linear Regression: The Basics

The goal is to find coefficients $\boldsymbol{\theta}$ that minimize the Residual Sum of Squares (RSS):

- **Model:** $\hat{y} = \theta_0 + \theta_1 x_1 + \cdots + \theta_p x_p = \boldsymbol{\theta}^T \mathbf{x}$ (with $x_0 = 1$ for the intercept).
- **Cost Function:** $J(\boldsymbol{\theta}) = \sum_{i=1}^{n} (y_i - \boldsymbol{\theta}^T \mathbf{x}_i)^2 = \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2$.

The Least Squares Solution (Normal Equations)

When the design matrix $\mathbf{X}$ has full column rank, the closed-form minimizer is:

$$\hat{\boldsymbol{\theta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}, \qquad \text{where } \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

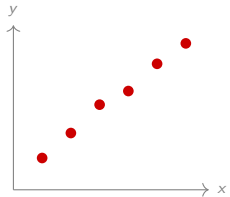
Perfect Fit Condition (Practice Paper 1, Q3.2)

If you have n distinct data points and fit a model with n parameters (e.g., a degree $n-1$ polynomial $\hat{y}_\theta(x) = \sum_{l=0}^{n-1} \theta_l x^l$), the curve interpolates every point and achieves zero training error—a classic recipe for **Overfitting**!

Worked Example: Linear & Polynomial Fit (Practice Paper 1, Q3)

Training Data ($n = 6$):

i	x_i	y_i
1	1.0	2.8
2	2.0	5.0
3	3.0	7.5
4	4.0	8.7
5	5.0	11.1
6	6.0	12.9



Questions & Step-by-Step Solutions:

- **(1) Least Squares Fit:** Fit $y = \theta_0 + \theta_1 x$ using:

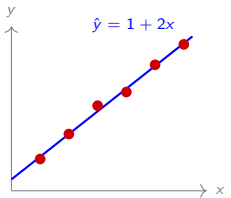
$$\mathbf{X}^T\mathbf{X} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} = \begin{bmatrix} 6 & 21 \\ 21 & 91 \end{bmatrix}, \quad \mathbf{X}^T\mathbf{y} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix} = \begin{bmatrix} 48 \\ 203 \end{bmatrix}$$

- **(2) 5th-Order Polynomial:** What training error $J(\hat{\theta})$ do we get with $\hat{y}_\theta(x) = \sum_{l=0}^{5} \theta_l x^l$?

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Solution ($\det(\mathbf{X}^T\mathbf{X}) = 6(91) - 21^2 = 105$):

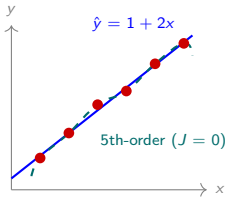
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- **(2) 5th-Order Polynomial:** What training error $J(\hat{\boldsymbol{\theta}})$ do we get with $\hat{y}_\theta(x) = \sum_{l=0}^{5} \theta_l x^l$?
Solution: $\hat{y}_\theta(x)$ has **6 parameters** $(\theta_0, \ldots, \theta_5)$ for **6 distinct points** $\implies$ interpolates every point with $J(\hat{\boldsymbol{\theta}}) = 0$ (**zero training error**).

Why Regularize?

Standard least squares can suffer from high variance when p is large or predictors are correlated. Regularization adds a penalty term to shrink the coefficients:

$$J_{\text{reg}}(\boldsymbol{\theta}) = \underbrace{\sum_{i=1}^{n}(y_i - \boldsymbol{\theta}^T \mathbf{x}_i)^2}_{\text{RSS (Fit to training data)}} + \underbrace{\lambda \cdot P(\boldsymbol{\theta})}_{\text{Penalty (Controls complexity)}}$$

- **Role of $\lambda \geq 0$ (Midterm 2022, Q4.3):**
  - **Small λ ($\lambda \to 0$):** Flexible model $\to$ Low Bias, High Variance (variance dominates).
  - **Large λ ($\lambda \to \infty$):** Heavily constrained $\to$ High Bias, Low Variance (bias dominates).
- **Intercept θ_0 is NOT penalized (Midterm 2022, Q1.6):** We do not want to shrink the baseline mean response of y.
- **Exam Questions (Practice Paper 1, Q1.5 & Q1.6):**
  1. *"Regularization decreases the bias of the model."*
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Ridge (L_2) vs. LASSO (L_1) Regression

Ridge Regression (L_2 Penalty)

- Penalty: $P(\boldsymbol{\theta}) = \sum_{j=1}^{p} \theta_j^2 = \|\boldsymbol{\theta}\|_2^2$.
- Shrinks coefficients smoothly toward zero as λ increases.
- Approaches zero asymptotically; does not perform feature selection (keeps all p predictors).

LASSO Regression (L_1 Penalty)

- Penalty: $P(\boldsymbol{\theta}) = \sum_{j=1}^{p} |\theta_j| = \|\boldsymbol{\theta}\|_1$.
- Shrinks coefficients and forces some to be exactly zero when λ is large enough.
- Performs automatic feature selection, yielding **sparse** models.

Applied Question (Practice Paper 1, Q3.3)

For the 5th-order polynomial fit on the 6 data points, match the three estimates to **Unregularized OLS**, **Ridge**, and **LASSO**:

- $\hat{\boldsymbol{\theta}}_a = [0.9,\ 2,\ 0,\ 0,\ 0,\ 0]^T$
- $\hat{\boldsymbol{\theta}}_b = [15,\ -29,\ 23,\ -7.9,\ 1.2,\ -0.07]^T$
- $\hat{\boldsymbol{\theta}}_c = [1.9,\ 0.44,\ 0.069,\ 0.01,\ 0.0014,\ 0.0002]^T$

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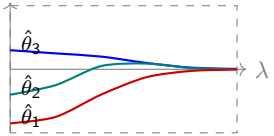
- $\hat{\boldsymbol{\theta}}_a = [0.9,\ 2,\ 0,\ 0,\ 0,\ 0]^T \longrightarrow$ LASSO: Sparse solution (high-order terms set to **exact zero**).
- $\hat{\boldsymbol{\theta}}_b = [15,\ -29,\ 23,\ -7.9,\ 1.2,\ -0.07]^T \longrightarrow$ Unregularized OLS: Huge oscillating coefficients to interpolate all 6 points.
- $\hat{\boldsymbol{\theta}}_c = [1.9,\ 0.44,\ 0.069,\ 0.01,\ 0.0014,\ 0.0002]^T \longrightarrow$ Ridge: All coefficients shrunk close to 0, but none are zero.

Interpreting Regularization Paths (Midterm 2022, Q4.1 & Q4.2)

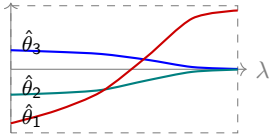
Match the three cost functions (for $\boldsymbol{\theta} = [\theta_1, \theta_2, \theta_3]^T$) to their coefficient paths **A**, **B**, and **C** as λ increases:

(i) $\text{RSS} + \lambda \sum_{i=1}^{3} \theta_i^2$ (ii) $\text{RSS} + \lambda \sum_{i=2}^{3} \theta_i^2$ (iii) $\text{RSS} + \lambda \sum_{i=1}^{3} |\theta_i|$

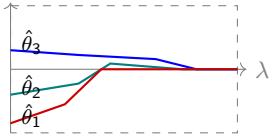
Graph A



Graph B



Graph C



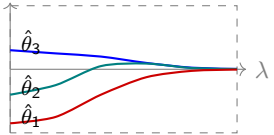
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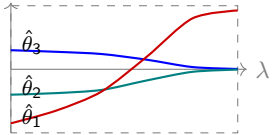
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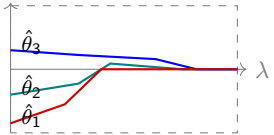
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Graph C



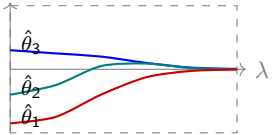
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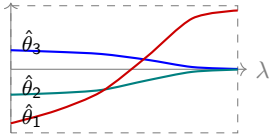
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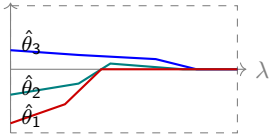
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- **Q4.2 (Shrink toward $\tilde{\theta} = [1, 1, 1]^T$ instead of 0):** Use penalty $\lambda \sum_{i=1}^{3} (\theta_i - 1)^2$.

Beyond Ridge & LASSO: Elastic Net

Exam Question (Midterm 2022, Q2.5)

"Briefly explain what is Elastic Net."

- **Definition (Macro):** A regularized regression method whose penalty is a convex combination of the LASSO (L_1) and Ridge (L_2) penalties:

$$P_{\text{ElasticNet}}(\boldsymbol{\theta}) = \alpha \sum_{j=1}^{p} |\theta_j| + (1-\alpha) \sum_{j=1}^{p} \theta_j^2, \quad \alpha \in [0,1]$$

- **Why use it? (Micro — Best of both worlds):**
  - Like **LASSO** (L_1), it produces sparse solutions and performs automatic feature selection.
  - Like **Ridge** (L_2), it handles highly correlated predictors effectively—shrinking correlated features together in groups rather than arbitrarily picking one and dropping the rest.

Evaluating Classifiers: Precision & Recall

Overall accuracy can be misleading when classes are imbalanced. Instead, focus on the positive class:

- **Precision:** Among all instances **predicted as positive**, what fraction is truly positive?
  Precision $= \frac{\text{TP}}{\text{TP}+\text{FP}}$
- **Recall (Sensitivity / TPR):** Among all **actual positive** instances, what fraction did we catch?
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*In spam detection ("spam" = positive), if you **raise the classification threshold** (e.g., from 0.5 to 0.8), what happens to recall and precision? Briefly explain.*

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- **Macro takeaway:** Predicting "spam" becomes more conservative $\Longrightarrow$ Recall decreases (or stays the same) and Precision probably increases.
- **Micro reason:** TP decreases/stays constant while FN increases/stays constant (missing more actual spam); keeping only high-confidence predictions reduces false positives (FP $\downarrow$).

Beyond Precision/Recall: The ROC Curve

The Receiver Operating Characteristic (ROC) curve evaluates a classifier across **all possible thresholds**:

- Plots True Positive Rate (TPR) vs. False Positive Rate (FPR) as the threshold varies from 1 to 0:

$$\text{TPR (Recall)} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \qquad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

- **Area Under the ROC Curve (AUROC / AUC):**
  - Summarizes threshold-independent ranking quality in a single number $\in [0, 1]$.
  - **Interpretation:** The probability that the classifier assigns a higher score to a randomly chosen positive instance than to a randomly chosen negative instance.
  - AUC $= 1$: Perfect classifier. AUC $= 0.5$: Random guessing (diagonal line).

When to use?

AUROC is the standard general-purpose ranking metric. For heavily imbalanced datasets (where TN is huge), also check the Precision-Recall Curve (AUPRC).

Linear vs. Quadratic Decision Boundaries

- **Logistic Regression:** Models $P(Y = 1 \mid \mathbf{x}) = \sigma(\beta^T \tilde{\mathbf{x}}) = \frac{1}{1+e^{-\beta^T \tilde{\mathbf{x}}}}$.
  - Decision boundary $P(Y = 1 \mid \mathbf{x}) = 0.5 \iff \beta^T \tilde{\mathbf{x}} = 0$ is **linear**.
  - **Softmax operator (Practice 1, Q2.4):** Maps real-valued multi-class scores into probabilities $(0, 1)$ summing to 1.
- **LDA vs. QDA (Midterm 2022 Q2.4, Practice 1 Q1.3):** Both assume $\mathbf{x} \mid Y = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$.
  - **LDA:** Assumes a common covariance matrix $\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}$ across all classes $\rightarrow$ Linear boundary. (LDA is a special case of QDA: TRUE!)
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Exam Questions (Practice Paper 1 Q1.2, Midterm 2022 Q1.2)

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   FALSE: Both have linear boundaries, but they are fitted differently (conditional MLE vs. Gaussian sample moments).

k-Nearest Neighbors (kNN)

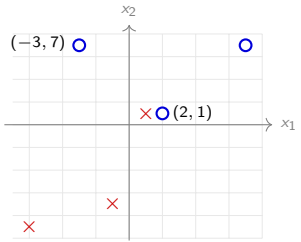
A classic non-parametric, instance-based discriminative classifier:

- **Algorithm:** Given a query point $\mathbf{x}_0$, find the K closest training points (by Euclidean distance) and predict the **majority class** among those K neighbors.
- **How K controls the Bias–Variance Tradeoff:**
  - Small K (e.g., $K = 1$): Very jagged boundary, **low bias, high variance**.
  - Key Fact: A **1-NN classifier always achieves 0% training error** (assuming distinct training inputs), because every training point's closest neighbor in the training set is **itself**!
  - Large K: Smoother boundary averaging over more neighbors $\rightarrow$ **higher bias, lower variance**.
- **Practical Note:** Because kNN relies on Euclidean distances, it is sensitive to feature scales (unlike Decision Trees).

Worked Example: Logistic, Tree & kNN (Midterm 2022, Q3)

Dataset ($n = 6$) & Visual Illustrations:

x_1	x_2	y	x_1	x_2	y
-6	-9	$\times$	-3	7	$\circ$
-1	-7	$\times$	7	7	$\circ$
1	1	$\times$	2	1	$\circ$



Questions & Step-by-Step Solutions:

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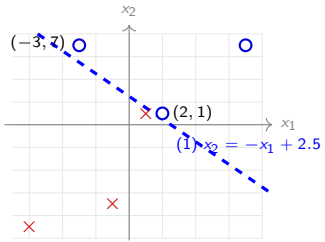
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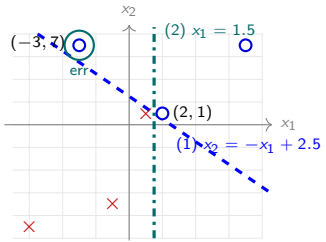
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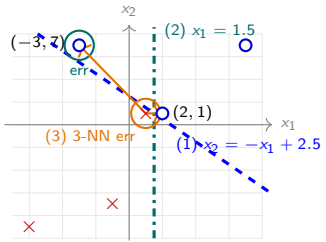
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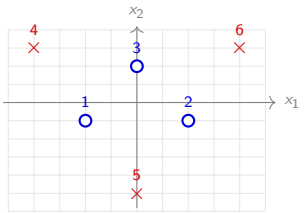
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   17% error (1/6): For $(1, 1)[\times]$, its 3 closest points are $(1, 1)[\times]$, $(2, 1)[\circ]$, and $(-3, 7)[\circ]$
   ($d = 0, 1, \sqrt{52} < \sqrt{68}$) $\rightarrow$ voted $\circ$!

Classifier Comparison: Problem Setup (Practice Paper 1, Q4)

Dataset ($n = 6$ observations):

Pt	x_1	x_2	y	Pt	x_1	x_2	y
1	-2	-1	$\circ$	4	-4	3	$\times$
2	2	-1	$\circ$	5	0	-5	$\times$
3	0	2	$\circ$	6	4	3	$\times$



Question (Practice Paper 1, Q4.2)

Six classifiers are trained on this 6-point dataset:

(i) Logistic Regression

(ii) LDA

(iii) QDA

(iv) kNN with $K = 1$

(v) kNN with $K = 3$

(vi) Classification Tree with a **single binary split**

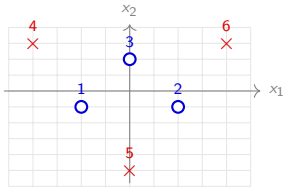
Match each classifier to its training misclassification error:

50% **33%** **33%** **17%** **0%** **0%**

Hint: Use decision boundary geometry—no heavy calculation needed!

Classifier Comparison: Step-by-Step Solution (Practice Paper 1, Q4)

Boundary Geometry on the 6 Points:



Q4.2 Summary Matching:

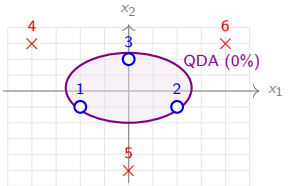
- 0% (0/6): **(iv) 1-NN**, **(iii) QDA**
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- 33% (2/6): **(i) Logistic**, **(ii) LDA**
- 50% (3/6): **(v) 3-NN**

Step-by-Step Visual Justifications:

1. **(iv) 1-NN (0%), (iii) QDA (0%) & (v) 3-NN (50%):**
2. **(vi) Single-Split Tree (17%) vs. (i)–(ii) Linear (33%):**
3. **Q4.3 (Move ONE point for linear separability):**

Classifier Comparison: Step-by-Step Solution (Practice Paper 1, Q4)

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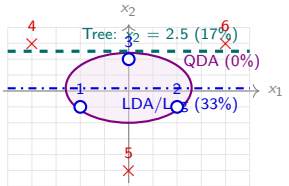
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- 33% (2/6): **(i) Logistic**, **(ii) LDA**
- 50% (3/6): **(v) 3-NN**

Step-by-Step Visual Justifications:

1. **(iv) 1-NN (0%), (iii) QDA (0%) & (v) 3-NN (50%):**
   - **1-NN (0%):** Every point's closest neighbor is itself (0/6).
   - **QDA (0%):** Purple quadratic ellipse encloses inner $\circ$ cluster $(1, 2, 3)$.
   - **3-NN (50%):** Inner $\circ$ $(1, 2, 3)$ are tightly clustered while outer $\times$ $(4, 5, 6)$ are far apart $\implies$ every point has ≥ 2 $\circ$ neighbors (predicts $\circ$ everywhere, 3/6).
2. **(vi) Single-Split Tree (17%) vs. (i)–(ii) Linear (33%):**
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Classifier Comparison: Step-by-Step Solution (Practice Paper 1, Q4)

Boundary Geometry on the 6 Points:



Q4.2 Summary Matching:

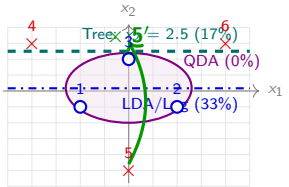
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Step-by-Step Visual Justifications:

1. **(iv) 1-NN (0%), (iii) QDA (0%) & (v) 3-NN (50%):**
   - **1-NN (0%):** Every point's closest neighbor is itself (0/6).
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   - Move **Point 5** $(0, -5)[\times]$ up to $\mathbf{5}'$ above Point 3 $(0, 2)[\circ]$ (green arrow) $\implies$ $x_2 = 2.5$ separates all $\times$ from $\circ$ (0% error for all except 3-NN)!

Decision Trees: Core Concepts & Impurity Measures

- **How they work:** Recursively split the feature space into axis-aligned rectangles R_m, predicting the majority class in each leaf.
- **Impurity Measures** (for node m with class proportions $\hat{p}_{mk}$, minimized at each split):
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1. **Practice Paper 1, Q1.9:** *"A classification tree with a single binary split is a linear classifier."*
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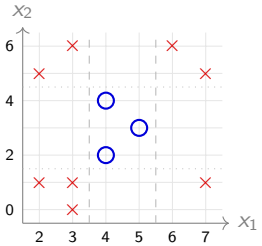
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Worked Example: Gini Impurity Split Calculation (Midterm 2022, Q5.1)

Dataset (11 points: 8 $\times$, 3 $\circ$):



Binary Gini & Split Criterion:

$$G_m = 1 - (\hat{p}_\circ^2 + \hat{p}_\times^2) = 2\hat{p}_\circ\hat{p}_\times$$

$$\text{Minimize } \frac{n_1}{n}G_1 + \frac{n_2}{n}G_2 \propto n_1 G_1 + n_2 G_2$$

Step 1: Evaluate the 4 candidate first splits:

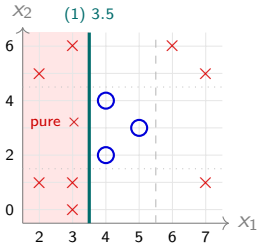
Split	n_1 $(\circ,\times)$	G_1	n_2 $(\circ,\times)$	G_2	$n_1G_1 + n_2G_2$
$x_1 < 3.5$	5 $(0,5)$	0	6 $(3,3)$	$\frac{1}{2}$	**3**
$x_1 < 5.5$	8 $(3,5)$	$\frac{30}{64}$	3 $(0,3)$	0	$\frac{30}{8} = 3.75 > 3$
$x_2 < 1.5$	4 $(0,4)$	0	7 $(3,4)$	$\frac{24}{49}$	$\frac{24}{7} \approx 3.43 > 3$
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- **1st Split:**
- **2nd Split (on right child $x_1 \geq 3.5$, with 3 $\circ$, 3 $\times$):**

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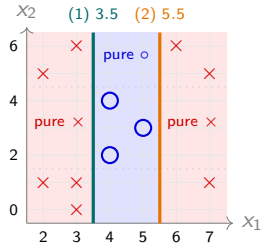
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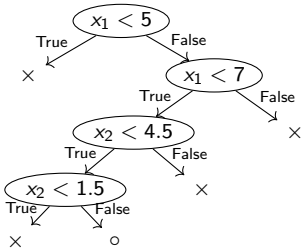
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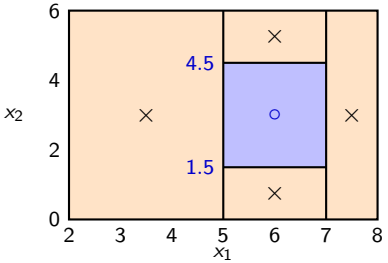
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- **2nd Split (on right child $x_1 \geq 3.5$, with 3 $\circ$, 3 $\times$):** Splitting at $x_1 < 5.5$ separates all 3 $\circ$ ($x_1 \in \{4,5\}$) from all 3 $\times$ ($x_1 \in \{6,7\}$) with **zero impurity**!

Visualizing a Tree's Decision Boundary (Midterm 2022, Q5.2 & Q5.3)

Given Tree (Midterm 2022, Q5.2):



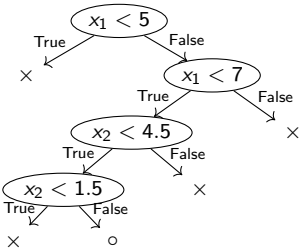
Induced 2D Feature Space Partition:



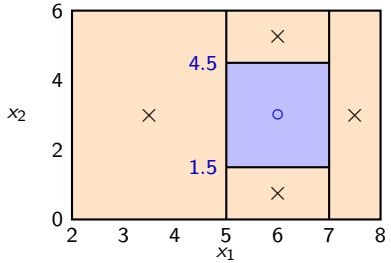
Midterm 2022, Q5.3: Three Non-Tree Models with 0% Training Error on Q5.1 Data?

Visualizing a Tree's Decision Boundary (Midterm 2022, Q5.2 & Q5.3)

Given Tree (Midterm 2022, Q5.2):



Induced 2D Feature Space Partition:



Midterm 2022, Q5.3: Three Non-Tree Models with 0% Training Error on Q5.1 Data?

Since the ○ cluster is surrounded by × (not linearly separable), linear classifiers fail, but 1-NN (kNN with $K = 1$), QDA, and Kernel SVM (with RBF kernel) all achieve **0% training error**!

Ensemble Methods: Bagging, Random Forests, Boosting

Key Idea

Combine many base learners (decision trees) into a single strong ensemble to improve test generalization.

Bagging & Random Forests

- Grow deep trees in parallel on **bootstrap samples** of the training data and average/vote.
- **Primary Goal:** Reduce Variance of high-variance base trees (Practice 1 Q1.10, Midterm 2022 Q1.9).
- **Random Forests** improves on Bagging by considering only a random subset of m predictors at each split, **de-correlating** the trees.
- *Exam Trap (Practice 2, Q1.3):* RF is an extension of **Bagging**, not AdaBoost!

Boosting (AdaBoost / Gradient Boosting)

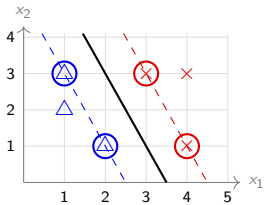
- Grow trees sequentially: each new tree focuses on the errors/residuals of previous trees.
- **Primary Goal:** Reduce Bias.
- *Exam Trap (Midterm 2022, Q1.10): "Boosting is typically used to improve large models with small bias and high variance."*
  $\rightarrow$ FALSE! Boosting uses **simple base models** (shallow stumps) with high bias and low variance.

Support Vector Machines (SVM): Key Ideas

- **Maximal Margin Classifier (Hard-Margin SVM):** For linearly separable data, finds the hyperplane $\mathbf{w}^T\mathbf{x} + b = 0$ that maximizes the margin (perpendicular distance to the closest points of either class).
  - Under canonical constraints $y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1$, the one-sided margin is $M = \frac{1}{\|\mathbf{w}\|}$ (total street width $\frac{2}{\|\mathbf{w}\|}$).
  - **Dimension of Hyperplane (Practice 2, Q4.3):** Depends only on the number of features p (a $p-1$ dimensional flat subspace in $\mathbb{R}^p$).
- **Support Vectors:** The critical training points that define the hyperplane:
  - In hard-margin SVM, they lie exactly on the margin boundaries $\mathbf{w}^T\mathbf{x} + b = \pm 1$.
  - Moving any non-support vector (without crossing the margin) **does not change** the boundary at all!
- **Soft-Margin SVM & Parameter `C` in `sklearn` (Practice 2, Q4.4):**
  - Objective: $\min_{\mathbf{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i$, where C is the cost (penalty) of margin violations.
  - **Large C:** Strict penalty on errors $\rightarrow$ narrower margin, lower bias, higher variance (risk of overfitting).

Worked Example: SVM Margin & Support Vectors (Midterm 2022, Q6)

6 Training Points (△ vs. ×):



- **(1) 4 Support Vectors:**
  △: $(1,3), (2,1)$; ×: $(3,3), (4,1)$.
- **(2) Boundary & Margin:**
  Line: $2x_1 + x_2 = 7$; $M = \frac{2}{\sqrt{5}}$ (width $\frac{4}{\sqrt{5}}$).

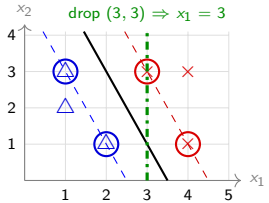
Q6.3 & Q6.4: What if we remove ONE support vector?

1. **Q6.3 (On this dataset):** Does the margin increase, decrease, or stay the same when we remove one SV?

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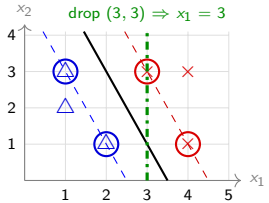
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2. **Q6.4 (On ANY dataset):** What happens in general when a support vector is removed? Why?
   Answer: Margin stays the same or increases—never decreases!
   • **Proof:** Hard-margin SVM is a **constrained maximization** problem where each point is a constraint; dropping a constraint enlarges the feasible set $(\max_{x \in A \cap B} f(x) \leq \max_{x \in A} f(x))$.

High-Frequency Exam Checklists & Additional Traps

Review All Lecture Topics & Quick Conceptual Checks:

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  - **Critically affects:** *k*NN & SVM (distances/margins depend on units) and Ridge / LASSO (penalties $\sum \theta_j^2$, $\sum |\theta_j|$ depend on coefficient scale).
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  1. **Q1.7:** *"Backward stepwise selection is guaranteed to yield the best subset of p predictors."*
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- **Stepwise Selection & GAMs (Midterm 2022, Q1.7 & Q1.8):**
  1. **Q1.7:** *"Backward stepwise selection is guaranteed to yield the best subset of p predictors."* $\rightarrow$ **FALSE** (both forward and backward stepwise are **greedy**; only Best Subset checks all 2^p models).
  2. **Q1.8:** *"GLM is a special case of GAM."* $\rightarrow$ **TRUE** (GAM replaces linear terms $\beta_j x_j$ with smooth functions $f_j(x_j)$; setting $f_j(x_j) = \beta_j x_j$ recovers a GLM).
- **Key `sklearn` Hyperparameters & Flexibility:**
  - Flexibility $\uparrow$ (Bias $\downarrow$, Variance $\uparrow$) when: `C` $\uparrow$ (SVM), `alpha`/λ $\downarrow$ (Ridge/LASSO), `max_depth` $\uparrow$ (Trees), `n_neighbors` $K \downarrow$ (kNN), Polynomial `degree` $d \uparrow$.

How to Prepare: A Summary

1. **Master the Core Tradeoffs & Distinctions:** Understand the Bias–Variance Tradeoff, Parametric vs. Non-Parametric, Generative vs. Discriminative, and the purpose of Regularization and Cross-Validation.
2. **Read True/False Statements Carefully:** Watch for absolute words (*"always," "guaranteed"*). Remember that "linear classifier" means a **linear decision boundary**, and non-parametric models do **not** always get 0% training error!
3. **Practice Hand Calculations:** Be comfortable computing 2×2 Normal Equations $(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$, linear boundary $\beta^T\tilde{\mathbf{x}} = 0$, Gini / Cross-Entropy splits, and SVM margins $\frac{1}{\|\mathbf{w}\|}$.
4. **Draw the Geometry:** Sketching the 2D points, hyperplanes, tree partitions, or regularization paths makes matching problems straightforward.
5. **Always Write Your Motivation:** On applied matching questions, a bare pairing without a short justification scores zero marks—state the geometric or algebraic reason clearly!

Good Luck on the Midterm!

Any Questions?